Hackathon Feature

Variables

Variable	Type	Visibility	Purpose
isApprovedHost	mapping(address => bool)	public	Registry of addresses admin-approved to create hackathons
maxXRDMPerParticipant	uint256	public	Global per-participant XRDM cap; 0 = disabled (unlimited)
nextHackathonId	uint256	public	Auto-incrementing counter for hackathon IDs (starts at 0, first hackathon = id 1)
hackathons	mapping(uint256 => Hackathon)	internal	Main storage — hackathon id → hackathon record

Struct

Field	**Type**	**Purpose**
host	address	Creator of the hackathon; address(0) means doesn't exist
isParticipant	mapping(address => bool)	Whether an address is a registered participant
isWinner	mapping(address => bool)	Whether an address is noted as a winner
xrdmAwarded	mapping(address => uint256)	Cumulative XRDM awarded per participant (for cap check)
nftCounter	uint256	Increments per NFT minted; ensures unique tokenIds

Functions - Admin(owner only)

Function	Signature	What it does
addHackathonHost	addHackathonHost(address host)	Marks host as approved to create hackathons; reverts if host is zero
removeHackathonHost	removeHackathonHost(address host)	Removes host's approval (idempotent — no error if not approved)
setMaxXRDMPerParticipant	setMaxXRDMPerParticipant(uint256 amount)	Sets global per-participant XRDM cap; 0 disables the cap

Functions - Host/lifecycle

Function	Signature	Caller	What it does
createHackathon	createHackathon() → uint256	Approved host	Creates a new hackathon; assigns next ID; caller becomes the host; returns the new hackathonId
registerParticipants	registerParticipants(uint256 hackathonId, address[] participants)	Host of that hackathon	Batch registers wallets as participants; idempotent; rejects zero addresses
awardXRDM	awardXRDM(uint256 hackathonId, address[] recipients, uint256[] amounts)	Host of that hackathon	Validates each recipient is a participant + amount > 0 + cap not exceeded; uses single batch mint for gas efficiency; emits per-recipient event
noteWinners	noteWinners(uint256 hackathonId, address[] winners)	Host of that hackathon	Batch marks participants as winners; additive across multiple calls; each must be a participant
awardNFT	awardNFT(uint256 hackathonId, address recipient, string uri)	Host of that hackathon	Mints a unique soulbound NFT to a participant with the given URI; auto-increments nftCounter for uniqueness

View functions

Function	Signature	Returns
getHackathonHost	getHackathonHost(uint256 hackathonId) → address	The host address; reverts HackathonNotFound if unknown
isHackathonParticipant	isHackathonParticipant(uint256 hackathonId, address addr) → bool	Whether addr is a registered participant; reverts if hackathon doesn't exist
isHackathonWinner	isHackathonWinner(uint256 hackathonId, address addr) → bool	Whether addr is a noted winner; reverts if hackathon doesn't exist
getXRDMAwardedInHackathon	getXRDMAwardedInHackathon(uint256 hackathonId, address addr) → uint256	Cumulative XRDM awarded to addr in this hackathon; reverts if hackathon doesn't exist